3.1. SPECIES OF ATOMIC NUCLEI

Both the protons and the neutrons in the nucleus are commonly denominated *nucleons*. The *mass number A* is the total number of nucleons. Thus

$$A = N + Z \tag{3.1}$$

where Z is the number of protons, i.e. the *atomic number* and N is the number of neutrons. The elemental identity and the chemical properties are basically determined by the atomic number even if most chemical interactions take place via the interactions of the surrounding electron cloud.

As we have seen in Chapter 1, an element may be composed of atoms that, while having the same number of protons in the nuclei, have different mass numbers and, therefore, different numbers of neutrons. Neon, for example, has an atomic number of 10, which means that the number of protons in the nuclei of all neon atoms is 10; however, 90% of the neon atoms in nature have 10 neutrons present in their nuclei while 10% of the atoms have 12 neutrons. Such atoms of constant Z but different A are called *isotopes*. The heavy hydrogen isotopes ^{2}H and ^{3}H are used so often in nuclear science that they have been given special names and symbols, deuterium (D) and tritium (T), respectively.

The word isotope is often misused to designate any particular nuclear species, such as ^{16}O, ^{14}C, ^{12}C. It is correct to call ^{12}C and ^{14}C isotopes of carbon since they are nuclear species of the same element. However, ^{16}O and ^{12}C are not isotopic since they belong to different elemental species. The more general word *nuclide* is used to designate any specific nuclear species; e.g. ^{16}O, ^{14}C, and ^{12}C are nuclides. The term *radionuclide* should be used to designate any radioactive nuclide, although, somewhat illogically, *radioisotope* is a common term used for the same purpose.

In addition to being classified into isotopic groups, nuclides may also be divided into groupings with common mass numbers and common neutron numbers. *Isotopes* are nuclides with a common number of protons (Z), whereas *isobar* is the term used to designate nuclides with a common number of nucleons (A), i.e. the same mass number. Nuclei with the same number of neutrons (N) but different atomic numbers are termed *isotones* ^{40}K and ^{40}Ar are examples of isobars, while ^{3}H and ^{4}He are examples of isotones.

In some cases a nucleus may exist for some time in one or more excited states and it is differentiated on this basis. Such nuclei that necessarily have the same atomic number and mass number are called *isomers*. ^{60m}Co and ^{60g}Co are isomers; the ^{60m}Co nuclide exists in a high energy (*excited*) *state* and decays spontaneously by emission of a γ-ray with a half-life of 10.5 min to the lowest energy, *ground state*, designated by ^{60g}Co.

$$^{60m}\mathrm{Co} \xrightarrow[10.5\ \mathrm{min}]{\gamma} {}^{60g}\mathrm{Co} \tag{3.2}$$

The symbol m stands for *metastable*, while g (or no symbol) refers to the ground state. In order for a nuclide to be called *metastable* one usually demand that the excited state has a half-live of at least 10^{-9} s. This is about 1000 times longer than for a normal excited

state to decay. There are a few nuclides having more than one metastable state. These are then designated as m_1, m_2, etc.

3.2. ATOMIC MASSES AND ATOMIC WEIGHTS

The *universal mass unit*, abbreviated u (sometimes amu for atomic mass unit), is defined as one-twelfth of the mass of the ^{12}C atom which has been defined to be exactly 12 u. The absolute mass of a ^{12}C atom is obtained by dividing the value 12 by the Avogadro number ($\mathbf{N}_A = 6.022137 \times 10^{23}$). The value for the mass of a ^{12}C atom, i.e. the nucleus plus the 6 extranuclear electrons, is thus $1.992\,648 \times 10^{-23}$ g. Atomic masses are expressed in units of u relative to the ^{12}C standard. This text uses M to indicate masses in units of u, and m in units of kilograms; $m = M/10^3\ \mathbf{N}_A$. One atomic mass unit corresponds to 931.5 MeV using the famous equation $E = m_0c^2$. Historically this has not always been the case. Up to as late as the 1960ies two different scales of atomic masses existed. The chemists used a scale based on the natural composition of oxygen and designated that to have a mass of 16 u while in the physics community the same mass was assigned to the most common oxygen isotope, ^{16}O. Since natural oxygen also contains the isotopes ^{17}O and ^{18}O this led to two different tables of atomic masses which naturally was highly inconvenient.

In nuclear science it has been found convenient to use the atomic masses rather than nuclear masses. The number of electrons are always balanced in a nuclear reaction, and the changes in the binding energy of the electrons in different atoms are insignificant within the degree of accuracy used in the mass calculations. Therefore the difference in atomic masses of reactants and products in a nuclear reaction gives the difference in the masses of the nuclei involved. In the next chapter, where the equivalence between mass and energy is discussed, it is shown that all nuclear reactions are accompanied by changes in nuclear masses.

The mass of the nucleus can be approximated by subtracting the sum of the masses of the electrons from the atomic mass. The mass of an electron is 0.000549 u which is equivalent to 9.1094×10^{-31} kg. Since the neutral carbon atom has 6 electrons, the approximate mass of the nucleus is $1.992\,648 \times 10^{-26} - 6 \times (9.1094 \times 10^{-31}) = 1.992\,101 \times 10^{-26}$ kg. This calculation has not included the difference in the mass of the 6 extra electrons attributable to the binding energy of these electrons. However, this binding energy has a mass equivalence which is smaller than the least significant figure in the calculation.

The mass of a neutron is 1.008 665 u while that of the hydrogen atom is 1.007 825 u. This shows the fact that the neutron decays to a proton and en electron. Since both neutrons and protons have almost unit atomic masses, the atomic mass of a nuclide should be close to the number of nucleons, i.e. the mass number. However, when the table of elements in the periodic system (Appendix I) is studied it becomes obvious that many *elements* have masses which are far removed from integral values. Chlorine, for example, has an atomic mass value of 35.453 u, while copper has one of 63.54 u. The variations in precision of the given atomic weights can be explained by the effect of the relative abundances of the stable isotopes of the elements contributing to produce the observed net mass.